

Dewell Guerand

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Education

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- MEng Double Degree**, CentraleSupélec, Université Paris-Saclay Aug 2026 – June 2028
- Engineering cycle – TIME Double Degree Network (Top European Engineering Exchange Programme)
- MEng**, Louvain School of Engineering (EPL), UCLouvain Sept 2025 – June 2026
- Major: Applied Mathematics
- BEng**, Louvain School of Engineering (EPL), UCLouvain Sept 2022 – June 2025
- Major: Applied Mathematics, Minor: Computer Science
 - Graduated *Cum Laude*
- Lycée de Berlaymont**, Belgium Sept 2016 – June 2022
- Graduated with honors – Major in Mathematics and Sciences

Experience

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- Quant Analyst**, LSM Investment Club – UCLouvain Oct 2025 – Present
- Conducted quantitative analysis and financial modelling to support investment decisions across equity and fixed income strategies.
 - Presented and popularized recent academic research in quantitative finance to club members; contributed to portfolio strategy discussions.

Projects

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- Interest Rate Modeling – Sinusoidal Hull-White Model** github.com/DewellGuerand
- Implemented and validated an interest rate simulation engine based on the Hull-White model, incorporating a sinusoidal mean-reversion extension derived from a research paper; calibrated model parameters and benchmarked against standard short-rate models.
 - *Tools: Python (NumPy, SciPy, Matplotlib)*
- Pricer for GMAB & GMDb Insurance Products**
- Developed a full pricing framework for Guaranteed Minimum Accumulation Benefit (GMAB) and Guaranteed Minimum Death Benefit (GMDb) contracts, covering closed-form analytical derivations and numerical methods including Binomial Trees and Monte Carlo simulation (Euler-Maruyama scheme); compared method accuracy and computational efficiency.
 - *Tools: Python (NumPy, SciPy)*
- Disease Detection – Machine Learning**
- Designed and trained a disease classification pipeline combining a CNN for image-based features with ensemble methods (XGBoost, RandomForest); performed hyperparameter tuning and cross-validation. *Tools: Python (TensorFlow, scikit-learn)*
- Finite Element Analysis – 2D Linear Elasticity**
- Developed a complete finite element analysis application for solving 2D linear elasticity problems in engineering structures, with mesh generation, solver, and visualization components.
 - *Tools: C, Python*

Personal Achievements

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- **Hockey:** 15+ years of competitive play at club level; coached and mentored junior players.
 - **Running:** Completed the Brussels 20 km (half-marathon distance), training consistently while managing a full academic workload.
 - **Market Investment:** Actively managing a personal portfolio; developed interest in equity markets, factor investing, and macroeconomic analysis.
 - **Senior Prom Trek:** Ranked 13th out of 120 High schools in the Walloon region in a 40 km endurance trek.

Skills & Languages

Programming: Python, C, C++, Java, Dart, Bash, Oz, Julia
Libraries: NumPy, SciPy, Matplotlib, Pandas, TensorFlow, scikit-learn
Tools: Git, Overleaf, Typst
Languages: French (C2, native) | English (C1, IELTS 7.0) | Dutch (B2)